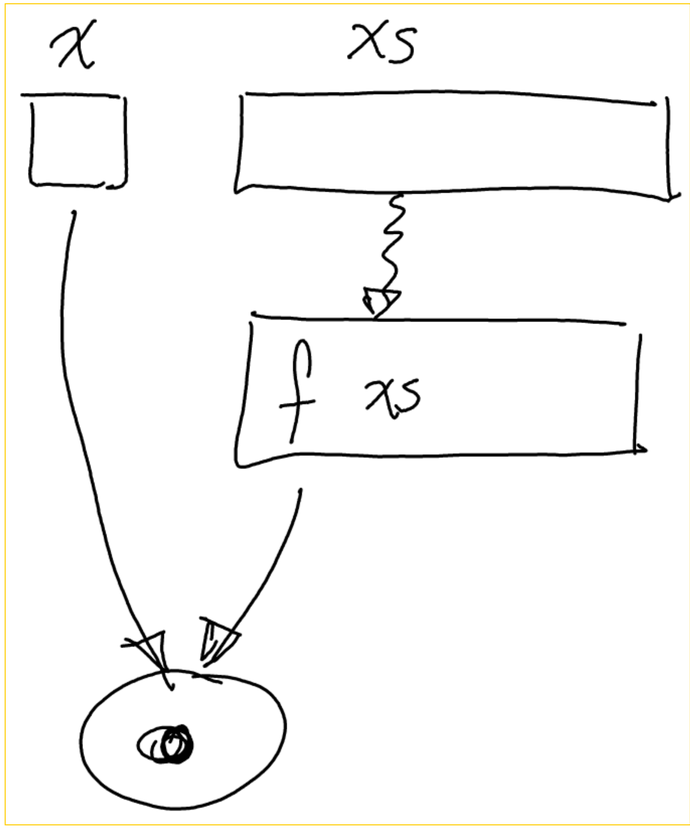
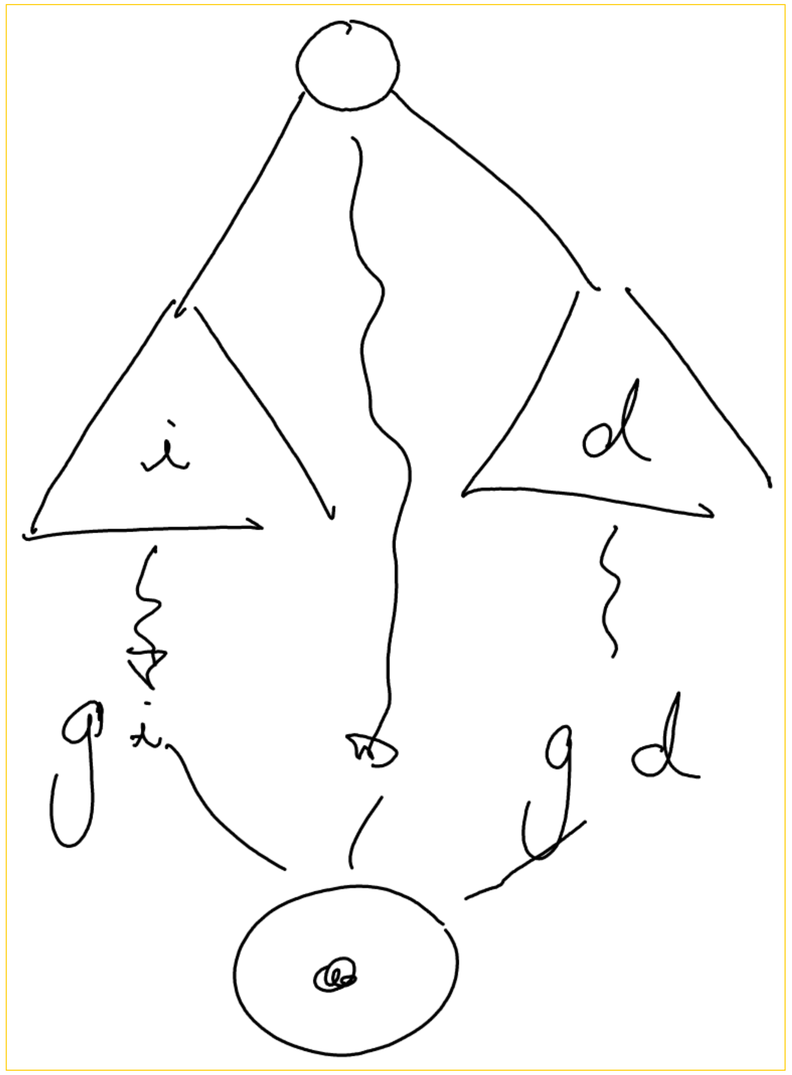
data [a] = [] | a : [a]

Nil | Cons a [a]





cantNodos :: AEB a → Int

cantNodos t = foldAEB

(\a → 1)

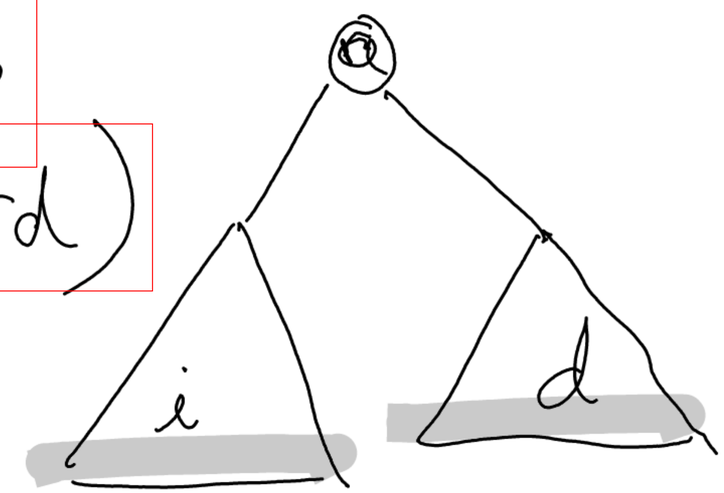
(\ri a rd → 1 + ri + rd)

(const 1)

t

Se pueden sacar.

cantNodos (Hoja 2) ⇝ 1

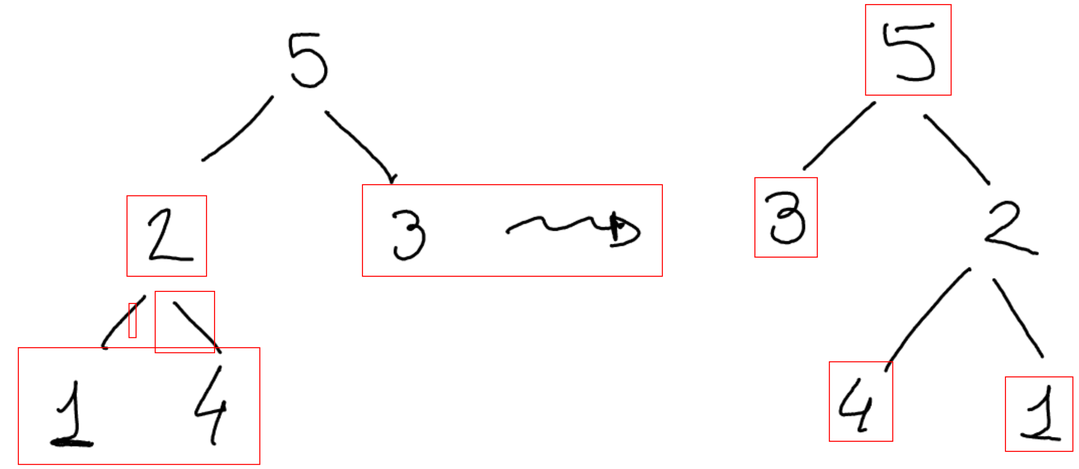


1 + cantNodos i + cantNodos d

cantNodos = foldAEB (const 1) (\i _ d → 1 + i + d)

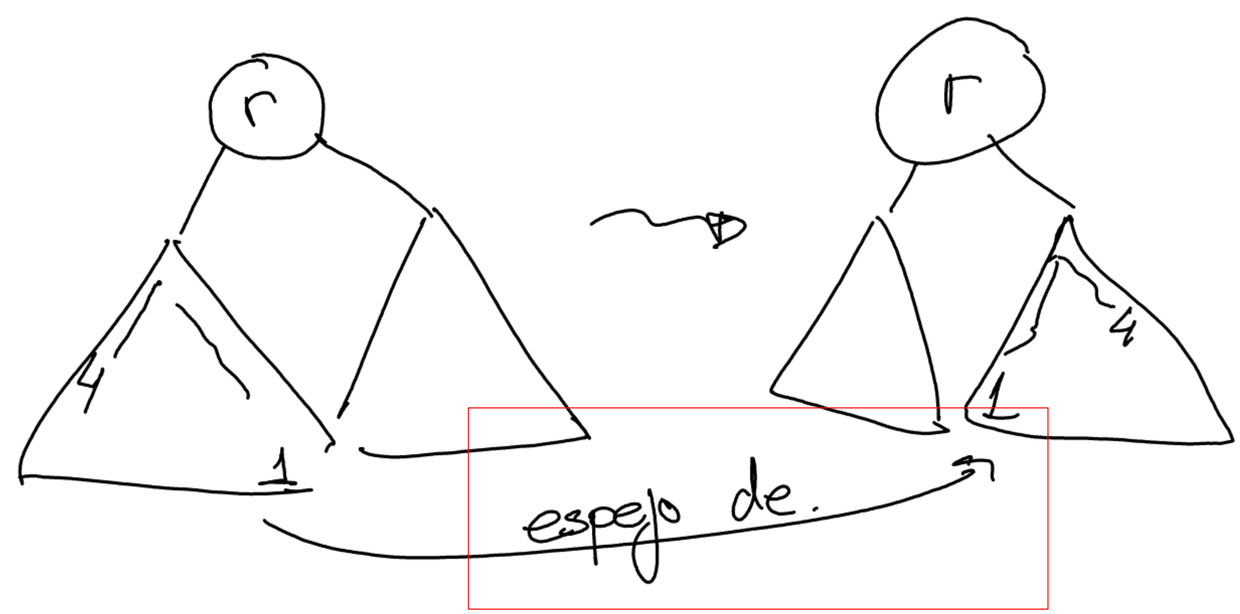
```
espejo :: AEB a → AEB a
espejo = foldAEB Hoja
         (\i r d →
           Bin d r i)
```

ya es el espejo del árbol izquierdo original.




```
Hoja :: a → AEB a
```

$\underbrace{\cancel{\lambda n \to \text{Hoja } n}}_{a \to AEB\ a}$



```
data AB a = Nil | Bin (AB a) a (AB a)

foldAB :: b -> (b -> a -> b -> b) -> AB a -> b

recrAB :: b -> (AB a -> b -> a -> AB a -> b -> b) -> AB a -> b

recrAB fNil fBin t =
  case t of
    Nil -> fNil
    Bin i r d -> fBin i (rec i) r d (rec d)
  where
    rec = recrAB fNil fBin
```

```
data Poli a = X | Cte a | Soma (Poli a) (Poli a) | Prod (Poli a) (Poli a)
```

```
evaluar :: (Num a) => a -> Poli a -> a
evaluar x X = x
evaluar x (Cte n) = n
evaluar x (Soma p q) = (evaluar x p) + (evaluar x q)
evaluar x (Prod p q) = (evaluar x p) * (evaluar x q)
```

$X^2 + 2X + 1 \equiv$ Soma (Soma (Prod X X) (Prod (Cte 2) X)) (Cte 1)

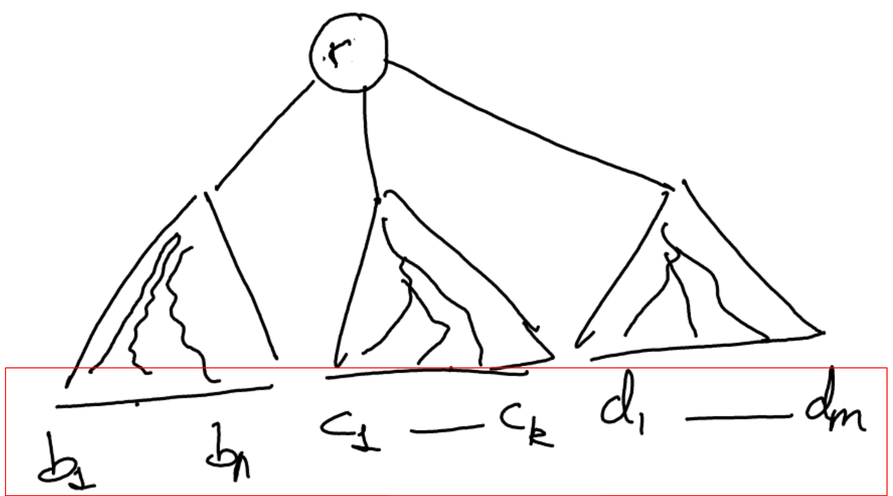
```
foldPoli :: b -> (a -> b) -> (b -> b -> b) -> (b -> b -> b) -> Poli a -> b
```

```
evaluar' x = foldPoli x id (+) (*)
```

```
data RoseTree a = Rose a [RoseTree a]
```

```
foldRT :: (a -> [b] -> b) -> RoseTree a -> b
foldRT f (Rose r ts = f r (map (foldRT f) ts)
```

```
ramas :: RoseTree a -> [[a]]
ramas = foldRT (\r res ->
    if null res then [[r]]
    else
        map (r:) (concat res)
)
```




```
(++) :: [c] -> [c] -> [c]
concat :: [[c]] -> [c]
concat res ~> [b1, b2 ... bn, c1 ... ck, d1 .. dm]
map :: (a -> b) -> [a] -> [b]
```

r : b1
⋮
r : bn
r : c1
⋮
r : ck
r : d1
⋮
r : dm

res [
[b1, b2 .. bn]
, [c1, c2 ... ck]
, [d1, ... dm]
]

bi :: [a]

[[a]]

res :: [[[a]]]